CYBERSECURITY PROJECT REPORT

NETWORK SCANNING REPORT

Localhost TCP Port Discovery and Security Observations

Target: 127.0.0.1 (loopback address)

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Assessment type: Controlled local TCP scan

Tools and method: Linux ss listener inventory and full-range TCP connect checks. Nmap reproduction commands are included for a Kali Linux rerun.

Scope note: 127.0.0.1 always refers to the machine where the scan command runs. The recorded results in this PDF came from the report-generation environment (Debian GNU/Linux 13), not from the user's Kali Linux computer. Re-run the commands on your own machine if your instructor requires machine-specific evidence.

1. Executive Summary

A local TCP port assessment was carried out against 127.0.0.1, the loopback interface of the machine executing the test. A full TCP connect check across ports 1-65535 identified eight stable listening TCP ports during the follow-up inventory. A transient response was also observed on port 51090 during the scan but was not present in the immediate follow-up listener snapshot. An HTTP request to port 8080 returned a 404 response and exposed a Uvicorn server header.

Method transparency: Nmap was not installed in the environment used to collect these results. The actual results were collected with a TCP connect check and verified against Linux `ss`. Nmap commands for reproducing the assessment on Kali Linux are provided in Section 7.

Assessment at a glance

Field	Recorded value
Target	127.0.0.1 (IPv4 loopback)
Environment	Debian GNU/Linux 13 report-generation environment; not the user's Kali host
Date	16 October 2026
Coverage	TCP ports 1-65535; local socket inventory via `ss -tuln`
Stable TCP listeners	8 observed in the follow-up `ss` snapshot
UDP assessment	No active UDP listener shown in the captured `ss -tuln` output; no UDP port scan performed

2. Objectives

• Identify TCP ports on the local loopback address that accept a connection.

• Compare connection-test results with the system listener inventory.

• Identify any service information visible without performing intrusive tests.

• Document findings, limitations, and reasonable defensive follow-up actions.

3. Scope and Authorization

In scope: TCP services listening on 127.0.0.1 of the environment where this report was generated.
Out of scope: External hosts, other users' devices, exploitation, password attacks, denial-of-service tests, and UDP scanning.
Authorization: Localhost-only testing in the execution environment. This report does not claim access to the user's computer.

4. Methodology

Step	Method	Purpose
1	`ss -tuln`	Record local listening TCP/UDP sockets and their bind addresses.
2	TCP connect checks across ports 1-65535	Check which local TCP ports accept connections.
3	HTTP HEAD request to 127.0.0.1:8080	Capture a basic HTTP response header; no exploitation attempted.
4	Follow-up `ss -tuln` snapshot	Compare detected ports with the sockets still listening at capture time.

5. Results and Evidence

The following eight TCP listeners were present in the captured `ss -tuln` output, and TCP connection checks to these ports succeeded. Service labels are reported conservatively; a port number alone does not prove which application is behind it.

Port	State	Bind address	Service identification	Observation
1384/tcp	Open	0.0.0.0	Unidentified	Listening on all IPv4 interfaces in this environment; service ownership not confirmed.
8080/tcp	Open	0.0.0.0	HTTP; Uvicorn header observed	HTTP HEAD / returned 404 Not Found. This is not proof of a vulnerability.
33443/tcp	Open	127.0.0.1	Unidentified	Loopback-bound listener; not ordinarily reachable through a host network interface.
45643/tcp	Open	127.0.0.1	Unidentified	Loopback-bound listener; owner not identified.
46501/tcp	Open	127.0.0.1	Unidentified	Loopback-bound listener; owner not identified.
51391/tcp	Open	127.0.0.1	Unidentified	Loopback-bound listener; owner not identified.
51513/tcp	Open	127.0.0.1	Unidentified	Loopback-bound listener; owner not identified.
56261/tcp	Open	127.0.0.1	Unidentified	Loopback-bound listener; owner not identified.

Transient observation: TCP port 51090 accepted a connection during the full-range scan, but it was not present in the immediately following `ss -tuln` output. It is therefore recorded as transient/unconfirmed rather than as a stable listener. Repeat the scan if this port is relevant to the lab.

6. Interpretation

Finding	Interpretation
Ports 1384 and 8080 bind to 0.0.0.0	A 0.0.0.0 bind address means the service listens on all local IPv4 interfaces. Actual reachability from other systems depends on routing, host/container isolation, and firewall rules.
Six stable ports bind to 127.0.0.1	These services are bound to loopback. They are normally accessible only from the same network namespace/host and are not directly exposed through its ordinary external interface.
Port 8080 returned HTTP 404	The HTTP service responded, and a Server header identified Uvicorn. A 404 response means the requested path was not found; it is not, by itself, a security issue.
No service owner confirmed for most ports	Identify owning processes and business purpose before deciding whether a listener is expected or should be disabled.

7. Risk Assessment and Recommendations

This scan did not confirm a vulnerability. Recommendations below are follow-up checks based on observed exposure, not claims that the services are insecure.

Priority	Observation	Recommended action
Review	TCP 1384 and 8080 listen on 0.0.0.0	Identify the owning process; confirm whether each service is required on every interface. Bind to loopback or restrict with firewall rules when appropriate.
Review	8080 responds with Uvicorn header	Confirm the application owner, version, intended exposure, authentication, and production configuration. Remove unnecessary server/banner details where appropriate.
Review	High-numbered loopback services are unidentified	Use process-level inspection to map each port to its process and purpose. Do not terminate services until the owner and dependency are known.
Monitor	Port 51090 was transient during scan	Repeat the scan and capture timestamps if reproducibility matters; transient application sockets can change during measurement.

8. Reproduce on Kali Linux

Run these commands on the actual Kali computer if you need evidence from your own machine. The results will likely differ because 127.0.0.1 refers to whichever machine runs the command.

```
sudo apt update && sudo apt install nmap
nmap -sT -sV -p- 127.0.0.1 -oN nmap-localhost.txt
ss -tuln | tee ss-listening.txt
sudo ss -lntup | tee ss-processes.txt
```

Open `nmap-localhost.txt` and replace the observed port table in this report if your assignment requires the results from your Kali installation. Keep the raw output as evidence.

9. Limitations

• These results are from the report-generation environment, not the user's personal Kali host.

• Nmap was unavailable during result collection; the actual measurement used TCP connect checks plus the Linux socket inventory.

• No UDP port scan, operating-system fingerprinting, authenticated inventory, vulnerability scan, or exploitation was performed.

• Service identification can be incomplete; a port number or HTTP header is not a complete software inventory.

10. Conclusion

The localhost assessment identified eight stable TCP listeners and one transient TCP response during a full-range connect check. Two stable listeners were bound to all IPv4 interfaces in the scanned environment, while six were bound to loopback. Port 8080 returned an HTTP 404 response with a Uvicorn server header. No vulnerability was confirmed. The next step is to identify process owners, validate whether each service is required, and reproduce the scan on the intended Kali machine if machine-specific submission evidence is required.

Evidence captured: Linux `ss -tuln` output; TCP connect checks across ports 1-65535; and an HTTP HEAD request to port 8080. Raw scan files should be attached when the test is repeated on Kali.